# COLREGS

What she is, what she is doing, and who gives way

**Version 1.15** — a training aid for the International Regulations for Preventing Collisions at Sea, offline, on a phone, a tablet or a terminal.

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## 1. What this is, and why it is built this way

COLREGS teaches the four things a watch leader has to produce in the dark, from what is in front of him:

1. **What she is** — from her lights at night, or her day shapes by day.
2. **What she is saying** — from her sound signals.
3. **Who gives way** — and what the action actually is.
4. **Where she is going** — from bearing drift, which is the one observation that answers it without a plot.

Four constraints shaped it, and each has a consequence worth knowing.

**1. It has to work with the antenna down.** Consequence: everything ships with the program. There is no lookup, no account and no year in which it stops working. The browser build caches itself on first visit and then needs nothing.

**2. It has to be usable at night without spoiling your night vision.** Consequence: a night mode in which every lamp goes red and the letters **W R G Y** beside each lamp carry the pattern. A dark-adapted eye costs about twenty minutes to recover from a bright panel, and that outranks fidelity of colour.

**3. It has to ask the question the way the sea asks it.** Consequence: nothing has a “show me the answer” button alone. Recognising a correct answer when it is put in front of you is far easier than producing one, and producing one is the job. Every view asks first and tells afterwards.

**4. The content must not be a retyped copy of anything.** Consequence: the browser build’s data was produced by running the shell engine’s own tables and printing what they built — twenty-eight vessels, ten day shapes, twenty-eight encounters, fifteen sound signals, fifteen lessons. The two builds cannot drift, because one is generated from the other.

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## 2. The two builds

The same content, in two places, because they are used at different moments.

| | the browser app | the shell tool |
|---|---|---|
| Where | `bashnav.navigationtools.org/web/colregs/` | `colregs` at a prompt |
| Lights, day shapes, sound signals | yes, drawn and sounded | yes, drawn in characters |
| Encounters | yes | yes |
| The fifteen rule lessons | yes | yes |
| Contact management | yes | yes |
| The collision-avoidance scenario | **no** — shell only | yes |
| Sound you can hear | **yes** | no: a terminal draws a blast |
| Reference tables, the card | by list in each view | `colregs ref`, `colregs card` |

**The browser build is the one to have on deck.** Add it to the home screen and it opens like an app, works with no signal, and draws the lamps as lamps.

**The shell tool is the one with the scenario** — the developing situation with a radar plot, three timed observations, the vector triangle and a CPA. It is the largest single piece in the program and it has not been rebuilt for the browser yet.

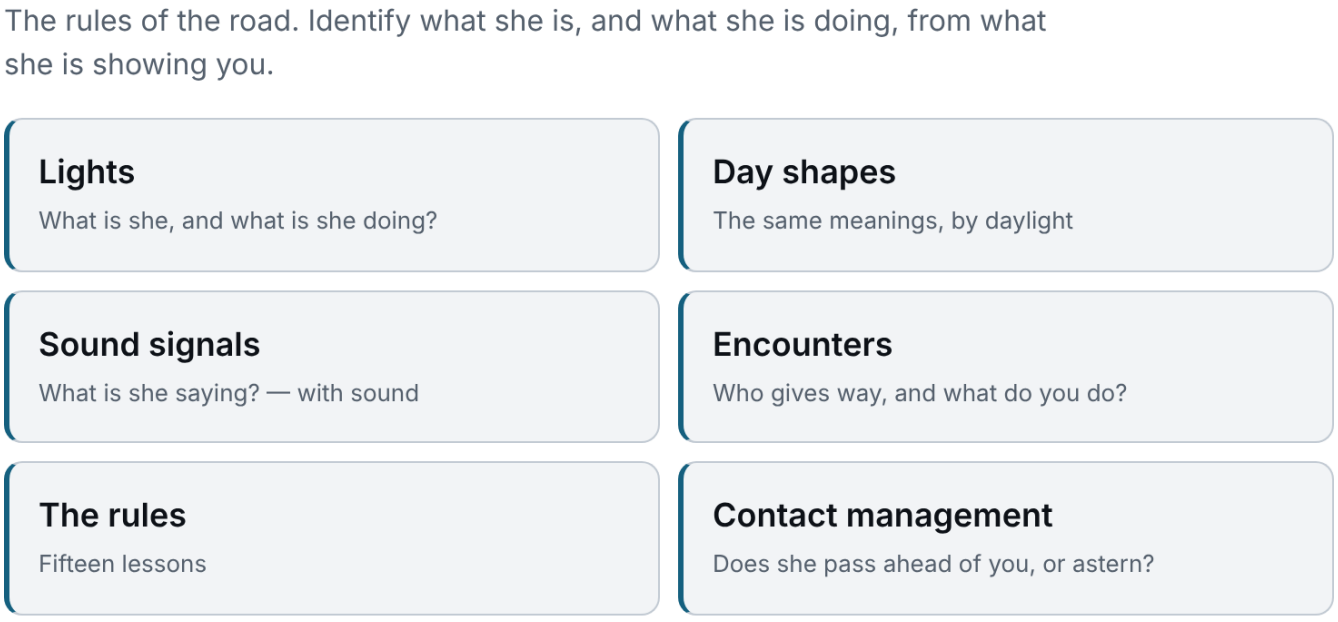



**Figure 1** — the browser app's six sections. Each one asks a question before it answers it.

## 3. Lights

A vessel is drawn from the bearing **you** are on, and you can move that bearing and watch her lights open and close.

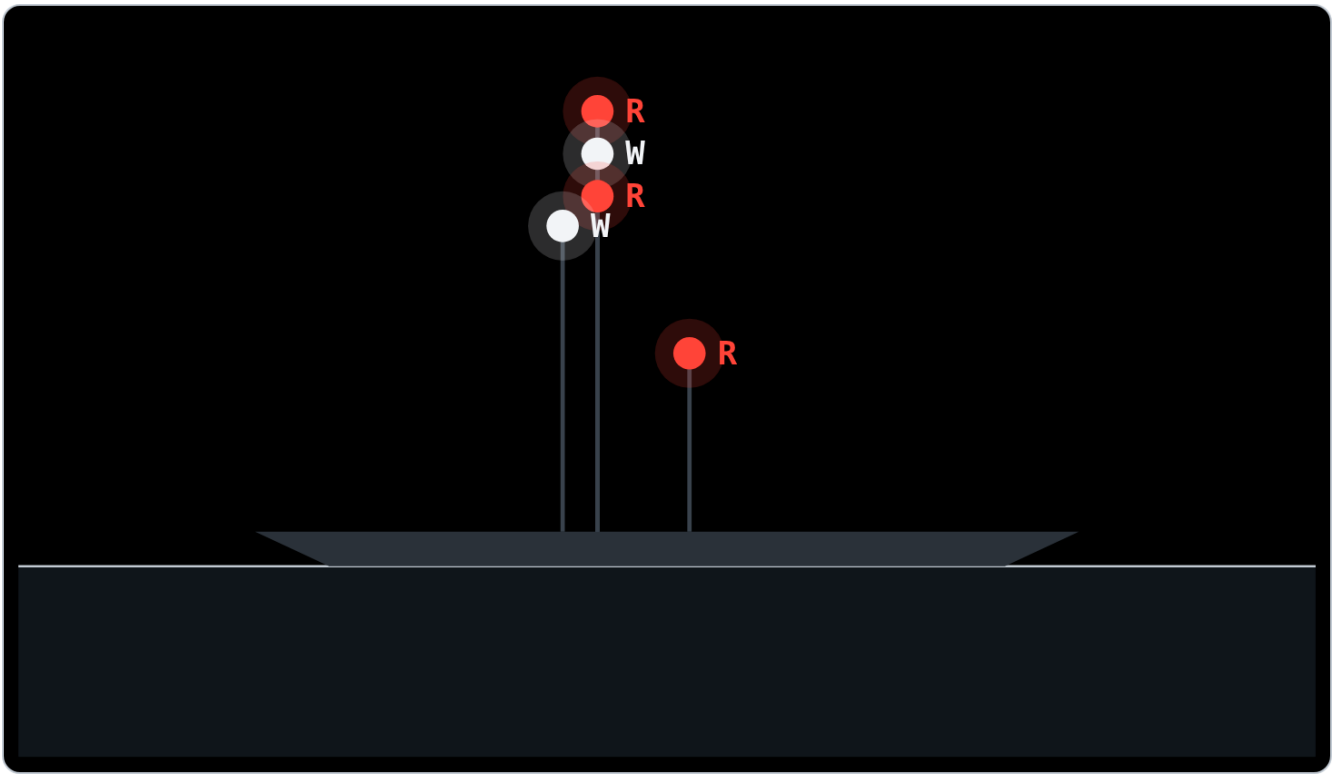



**Figure 2** — restricted in her ability to manoeuvre, Rule 27: red, white, red in a vertical line, and because she is making way, the ordinary lights as well. You are broad on her port bow, so her port sidelight shows and her starboard one does not.

### What the drawing is telling you, and what it is not.

1. **The lamps are where the rules put them** — the vertical order and the sectors are from the Convention, drawn rather than described.
2. **Sideways spacing is where a lamp sits along the hull,** not a heading cue. At any real range the lamps of one vessel are in line; the spread in the drawing exists so you can see which lamp is which.
3. **The sidelight cut-off is ±2.5°, not exact.** Annex I allows the cut-off to fall up to 3° outside the prescribed sector, so there is a small arc right ahead where **both** sidelights are seen. The drawing shows that, because a drawing that claimed a clean cut-off would teach you to call dead-ahead when you are not.
4. **Every lamp carries its letter** — W white, R red, G green, Y yellow. The pattern has to read with no colour at all, because in night mode it has none, and because colour is the first thing a tired eye gets wrong.
5. **A lamp marked `fl` flashes.** Annex I puts that at 120 or more flashes a minute, and the rate is part of the identification.

**Deck lights.** The **Deck lights off / on** button lights her working decks. Rule 20(b) allows working lights so far as they cannot be mistaken for the prescribed lights; a vessel at anchor may light her decks, and one of 100 m or more must (Rule 30(c)). Turn them on and see why a trawler working with her decks blazing is so much harder to read than the diagram suggests.

a) Restricted in her ability to manoeuvre

b) Fishing other than trawling, gear out over 150 m

c) Towing, tow over 200 m

d) Constrained by her draught

e) Dredging or underwater operations, restricted

*f) Show me what she is*

**Figure 3** — the question. Five candidates and a sixth row that gives up.

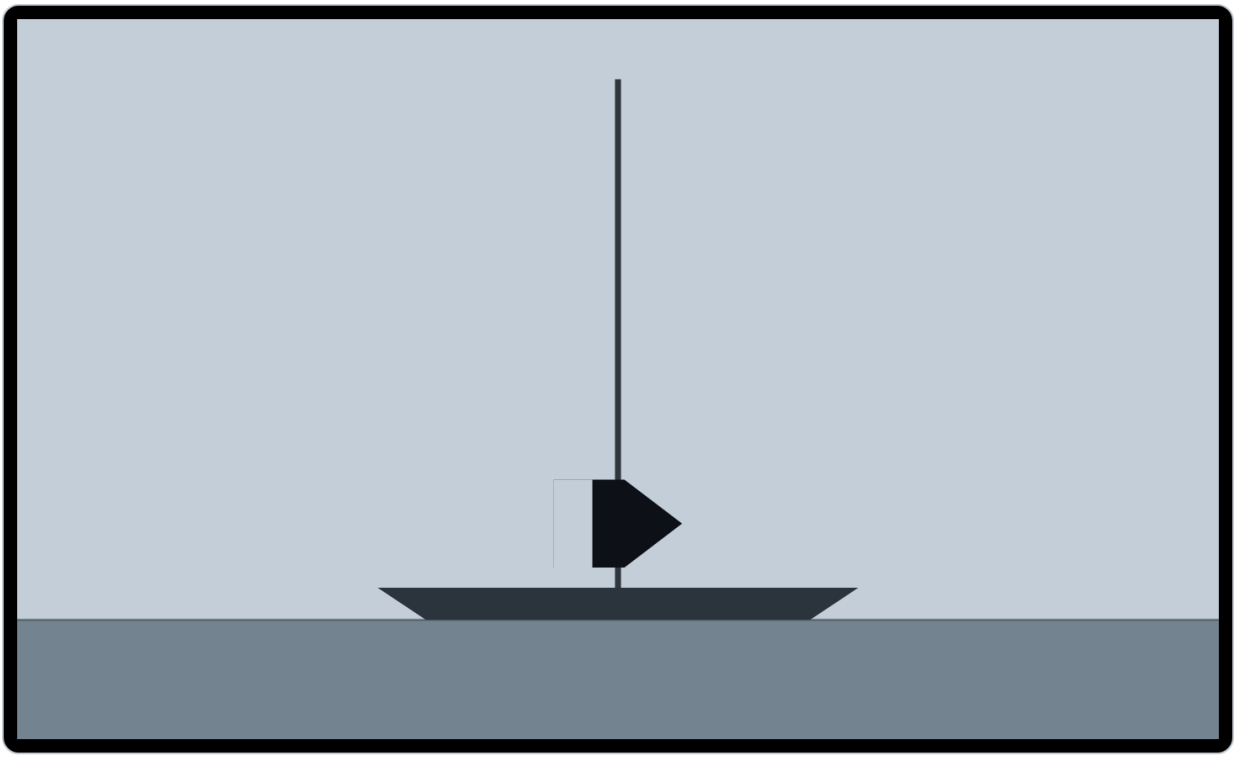
**The four wrong answers are chosen to be confusable with the right one.** Each candidate is scored on how much of its signature it shares with the answer — lamp colours for lights, the glyph stack for shapes, the sequence of blast kinds for sounds — and the closest four are offered, with a little noise so the same question is not always the same five rows. Over the whole data set the candidates offered share 3.79 lamp colours with the answer against 2.74 for random picks. **A trawler is set against a mine clearance vessel and a pilot vessel on duty, not against a seaplane.**

A wrong tap marks that row and leaves the others live, so a miss costs you another look at the picture rather than ending the question.

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## 4. Day shapes

The same meanings by daylight: ten of them, from the single black ball of a vessel at anchor to the ball-diamond-ball of a vessel restricted in her ability to manoeuvre.



Day shapes are **black**, seen against the sky. In night mode they go red with everything else — a bright panel costs you twenty minutes of dark adaptation, and that outranks fidelity.

**What is she?**

a) Vessel fishing

b) Vessel with divers down

c) Constrained by her draught

d) Vessel at anchor

e) Sailing vessel also under power

**Figure 4** — day shapes are black, against the sky.

**One deliberate inversion.** Day shapes are black. Drawing them white on a night sky would be backwards, so in dark mode this one view gets a daylight sky and black shapes. **In night mode it stays red on black** — fidelity loses to dark adaptation, as it should.

## 5. Sound signals

Fifteen signals, with the rule each comes from and the circumstances it is made in. The browser build makes the sound: short blasts, prolonged blasts and the bell, synthesised at their real durations.

A terminal could only draw a blast. This can make one.




▶ Sound it | Another

**What is she saying?**

a) A vessel being towed, and manned

b) Power-driven vessel making way through the water

c) Not under command, restricted in ability to manoeuvre, constrained by draught, sailing, fishing, or towing or pushing

d) I agree: go ahead and overtake

e) I am approaching a bend where other vessels may be hidden

f) *Tell me what she is saying*

# short, about a second  #### prolonged, four to six  * a stroke on the bell

### All fifteen, as a list

**Figure 5** — the blast pattern, drawn along a time axis.

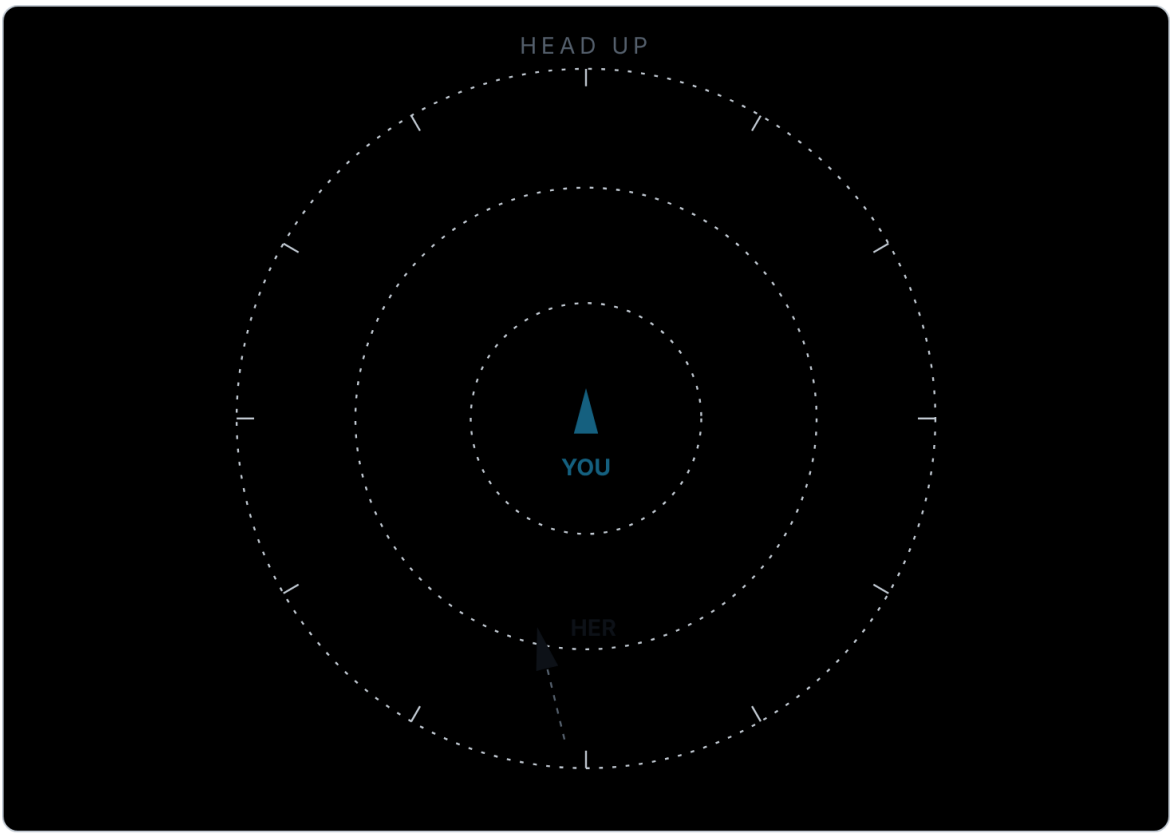
**A prolonged blast is supposed to feel long.** The shell tool’s engine says, in a comment, that a script cannot make a noise so it draws one. A browser can, and the difference matters: the distinction between one short blast and one prolonged blast is a distinction in duration, and reading it off a page teaches you the wrong thing.

The question is the same shape as the others — what is she saying, five candidates, one row that gives up.

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## 6. Encounters

Twenty-eight situations: what you are, what you see, and where she is.




You are a power-driven vessel making 8 knots.

You see a large power-driven vessel coming up astern, fine on your port quarter, heading as drawn.

a) Stand on: keep your course and speed.

b) Give way: alter to starboard.

c) Give way: slow down and let her pass.

d) Alter to port to open the distance.

**Another**

**Figure 6** — a head-up plot. You are at the bottom, heading up the page.

The plot is **head-up**: your own head is up the page, she is drawn where she bears and heading as she heads. Four answers are offered, and they are not “stand on / give way” alone — each names the **action**, because the rules require an action and not a category:

> *Give way: alter to starboard and pass down her port side.*

The wrong answers are the plausible ones: altering to port in a crossing situation, standing on when you are the give-way vessel, assuming a restricted vessel will keep clear of you when she cannot.

**Right and wrong carry a tick or a cross,** not only a green or red border. In night mode every colour token collapses to red, so a border alone would make the two states identical exactly where it matters most.

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## 7. The fifteen rule lessons

Each lesson is a short passage, a question and the reason behind the answer. Progress is kept, so you can do one at a time.

# L1 Rule 5 - Look-out

Every vessel shall at all times maintain a proper look-out by sight and hearing as well as by all available means appropriate in the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision.

1. It is the first substantive rule, and it is the one most often broken. More collisions are put down to a failure of look-out than to any other cause.
2. 'By sight and hearing' means a person, outside, looking and listening - not only a radar screen.
3. 'All available means' does include radar, AIS and VHF, and if you have them and do not use them properly, that is a breach too.
4. 'At all times' has no exceptions for singlehanders, for tiredness, or for the middle of an ocean.

The look-out must be able to appraise the situation. Someone at the wheel whose forward view is blocked by the sprayhood is not, by themselves, a proper look-out.

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Q. Which of these on its own satisfies Rule 5?
a) a radar with ARPA, watched from the chart table
b) AIS with a CPA alarm set  c) none of them

**a)**

**b)**

**c)**

**Figure 7** — a lesson. Read, answer, then the reason.

| | |
|---|---|
| L1 | Rule 5 — Look-out |
| L2 | Rule 6 — Safe speed |
| L3 | Rule 7 — Risk of collision |
| L4 | Rule 8 — Action to avoid collision |
| L5 | Rule 9 — Narrow channels |
| L6 | Rule 10 — Traffic separation schemes |
| L7 | Rule 12 — Sailing vessels |
| L8 | Rule 13 — Overtaking |
| L9 | Rule 14 — Head-on situation |
| L10 | Rules 15 and 16 — Crossing, and the give-way vessel |
| L11 | Rule 17 — Action by the stand-on vessel |
| L12 | Rule 18 — Responsibilities between vessels |
| L13 | Rule 19 — Restricted visibility |
| L14 | Rules 20 to 31 — Lights and shapes |
| L15 | Rules 32 to 37 — Sound and light signals |

In the shell tool, `colregs lesson L7` opens one directly, and `colregs numbers` drills which rule is which, both ways round.

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## 8. Contact management

The section a tracking party would recognise: what a contact's **bearing drift** tells you, how to say it, and the arithmetic you do in your head.

[ ] C1 Relative motion - the only plot that answers the question

[ ] C2 Drift toward the bow, and drift away from it

[ ] C3 Constant bearing, decreasing range

[ ] C4 The report

[ ] C5 The arithmetic you do in your head

[ ] C6 Ekelund - range out of a course change

[ ] C7 Relative bearings - so he can find her

0 of 7 read.

## Practice

Tracking party You have the plot. Which way, where, how close?

Ekelund Range out of a course change, with bearings only

The card The whole module on one screen

## Relative bearings, said as

`Red / Green` `Port / Starboard` `000 relative` `in words` `true bearing only`

**Figure 8** — seven lessons, and two exercises that mark their own answers.

| | |
|---|---|
| C1 | Relative motion — the only plot that answers the question |
| C2 | Drift toward the bow, and drift away from it |
| C3 | Constant bearing, decreasing range |
| C4 | The report |
| C5 | The arithmetic you do in your head |
| C6 | Ekelund — range out of a course change |
| C7 | Relative bearings — so he can find her |

### The rule the section is built on

> **Drifting away from your bow, she passes astern or never crosses your track at all. Drifting toward your bow, she crosses ahead — or draws ahead and clears.**

**It is not a rule of thumb.** Relative motion is a straight line, so the bearing sweeps one way and can never reverse: a contact drawing away from the bow cannot come back to it. It was checked over 700,000 random geometries — own speed 4 to 25 knots, any contact course and speed, range 2 to 15 miles — with no exception in either sector.

**The asymmetry is why it is worth saying out loud.** Drifting away, the miss is behind you and growing. Drifting toward, the miss is being built ahead of your bow, and it shrinks to zero as the drift rate does.

### Where good drift is still not safe

Rule 7(d)(i) is the constant-bearing case everyone knows. Rule 7(d)(ii) is the one that catches people, and C3 covers three of them:

1. **A very large vessel.** You plotted her bridge. Her bow is a ship's length nearer, and it is the bow that arrives.
2. **A tow.** The tug clears. The barge on the wire behind her does not.
3. **Close range.** Drift goes as the square of the range, so at close quarters a safe-looking drift is a small miss.

### The report

Designation, bearing, drift, range, CPA, time — **bearing before drift**, and **"steady" said out loud** rather than left as a silence that could mean you have not looked.

### The arithmetic

The three-minute rule, the six-minute rule, time to CPA, and one degree at one mile ≈ 35 yards for estimating the miss.

### Ekelund ranging — range from a change of course

Derived rather than quoted. The target's speed across the line of sight is unchanged between the two legs, so all of the change in bearing rate belongs to own ship:

```
R(kyd) = 1.91 × ΔS⊥(kt) / Δω(°/min)        S⊥ = own speed × sin(course − bearing)
```

Checked against the true range over 59,000 geometries to $8\times10^{-14}$ relative error, and over 300 generated leg pairs the worst recovery is 5.9% — which is bearing-rate rounding to a tenth of a degree a minute, not the formula.

### The two exercises

**`track` — the tracking watch.** Own course and speed, night orders of 3,000 yards, five marks three minutes apart. Three questions: the drift, where she goes, the CPA band. **One run in five is a genuine collision.** The reveal draws the relative-motion plot head-up with the CPA marked, and gives the report you would have made.

**`ekelund.`** Two legs with their bearing rates; name the range band. The reveal shows the working.

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## 9. The collision-avoidance scenario (shell only)

`colregs scenario` runs a situation as it develops: the radar plot, three timed observations, the vector triangle, the CPA, and then the true-motion animation of what actually happened — including what happens if you do nothing. At the lights prompt, `y` shows what **each** boat sees of the other, which is the part that explains why she is not altering.

It is the one section that is not in the browser build. Everything it needs is in the shell tool today; nothing in the browser build depends on it.

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## 10. Reference, and how a bearing is spoken

**Reference tables** — `colregs ref`: all the lights as a list, all the day shapes, and the sound-signal table. In the browser, each view has the same list folded away at the foot of it.

**The contacts card** — `colregs card`: the whole of section 8 on one screen, to read before a watch.

**Reporting style** — `colregs style <name>`, or `s` from the menu. It changes how a relative bearing is *said*; the true bearing is always given as well, because that is what goes on the chart and what agrees with the radar.

| | | |
|---|---|---|
| `rn` | Red and Green | *"bearing 311, Red 20, drawing left"* — Royal Navy and British practice. The side is carried by a **word**, which survives a bad intercom, a following sea and a tired listener. |
| `usn` | Port and starboard | The same information in plainer words; common in US and merchant practice. Longest to say, hardest to misunderstand. |
| `rel360` | Relative, full circle | *"340 relative"* — neat on a plot, worse in the dark: the listener has to work out which side 340 is on, which is the sum you were trying to save him. |
| `words` | Words only | *"fine on the port bow"* — slowest to say, fastest to act on. |
| `none` | True bearing alone | |

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# 11. The display modes

## The display: day, dark and night

The browser build has three display modes. They are not decoration: one of them exists to protect your night vision, and choosing the wrong one at the wrong moment costs you something real.

| Mode | What it is for |
|---|---|
| **NIGHT** | Red on black, for a dark-adapted eye. The default. |
| **DARK** | Near-black with light text, for below decks, dusk, or indoors. |
| **DAY** | White with near-black text, for daylight and for printing. |

### Why night is the default

A tool you open on deck in the dark must not start bright. A bright panel costs a dark-adapted eye roughly **twenty minutes** to recover from, and twenty minutes is a long time to be unable to see a mark or a ship’s lights properly. So the first time you open any of these tools you get night, and from then on you get whatever you last chose.

### Three buttons, not a cycle

The three modes are three separate buttons, and the one you are in is shown as selected. That is deliberate.

1. **A cycle can take you somewhere you did not intend.** One stray press with a cycling control can land you on DAY at three in the morning.
2. **A cycle cannot tell you where you are**, only what the next press would do. Three buttons show the current mode at a glance without changing it.
3. **Going darker is always one tap.** DARK and NIGHT are immediate from anywhere, because going darker never costs you anything.
4. **Going from NIGHT to DAY takes a deliberate half-second press.** Hold the DAY button and it fills as it goes; let go early and nothing happens. Nothing bright appears on the screen until the change actually completes. That one transition is guarded because that one transition is the only one that costs you your dark adaptation.

### The setting follows you

The mode is remembered for all of the tools together, and for the course pages that embed them. Set night in one and you are in night in the next. It is remembered in the browser, so it survives closing the app.

### It is right before the page draws

The mode is set in the first instructions the page runs, before any of it is painted. This matters on a first visit over a poor connection: set it later and the page would appear in one palette and then jump to another, which at night is the difference between keeping your

night vision and losing twenty minutes of it. If nothing has been stored yet, or whatever was stored cannot be read, the page opens in **night**.

### The controls follow the palette too

Sliders, scroll bars and the other controls the browser draws are told which world they are in, so they are drawn in the mode you chose rather than the browser's own light default. Without that, a night screen can carry a bright white slider across the middle of it.

### If you use High Contrast Mode

On Windows, High Contrast replaces every colour on the page with the ones your system theme specifies. That applies here too: the tools do not override it, because it is your setting and you chose it.

Nothing is lost by it. Where a colour carries meaning — the navigation lamps in colregs — the letter beside it says the same thing, so a lamp is still identifiable when the hue has gone.

**One thing to know if you need both.** High Contrast wins over night mode. If you need night vision on deck *and* high contrast to read, choose a dark high-contrast theme in Windows; the tools will follow it.

### What the display can and cannot do in sunlight

Worth knowing before you blame the app on a bright day.

In sunlight, light reflecting off the glass lands on the dark parts of the screen and the light parts alike. That lifts everything together and flattens the difference between them, so the contrast you actually see collapses no matter what colours the page uses:

| Condition | Body text, as seen |
|---|---|
| Indoors | about 10:1 |
| Overcast on deck | about 6:1 |
| Bright | about 3.5:1 |
| Full sun on the glass | about 2.3:1 |

**At the extreme, every colour scheme converges.** Pure black on pure white measures the same 2.3:1 in full sun as the day mode does, because the glare swamps the difference. A different palette is not the answer, and nor is a different shade of white: across every plausible choice of background the contrast changes by under 4%, which is below what an eye can see.

**What does work, in order:**

1. **Turn the screen brightness up.** This is the one large lever, and it is the device's, not the app's.
2. **Shade the screen** with a hand, your body or the sprayhood. Removing the reflection is worth more than any change the app can make.
3. **Use DAY mode**, which at least puts the maximum luminance the panel can give behind the darkest text it can draw.

### A note for printing

Documents and their figures are made in **day** mode. A screen captured in dark or night mode prints as a solid block of toner and loses the fine type in a working, so the printed pages show the day display even if that is not the one you use at sea.

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## 12. Offline, and putting it on the home screen

The browser build caches itself on the first visit — cache-first, because nothing in it changes while you are offshore and a lookup you cannot satisfy is only latency.

1. Open it once with a signal.
2. **Share → Add to Home Screen.**
3. Open it from the icon. It runs with the antenna down, full screen, with no browser furniture.

Opened straight from a file it also works — the content is in a script rather than fetched — but there is no cache to install and no icon.

The foot of the home screen says what it actually holds, so you can see whether it is installed before you need it to be.

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## 13. Where the content comes from

1. **The rules are the IMO's.** The Convention is what governs at sea. This program paraphrases to teach; where the two differ, the Convention is right.
2. **The browser build's data is generated from the shell engine**, by running its own table initialisers and dumping what they built. It is never hand-edited: it is regenerated, so the two builds cannot drift.
3. **The geometry is ported function for function** — the visibility arc, the bearing normalisations, the projection, the hull half-width — with the same names, so the awk and the JavaScript can be read side by side. Both are then run over all 361 integer bearings and the output compared. It is identical.
4. **That check earned its keep immediately.** The first port used a sidelight cut-off from an older copy of the engine. The diff caught it on four bearings — the four where the difference shows.

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## 14. Cautions

1. **This is a training aid, not the Convention.** The rules are published by the IMO and are what govern at sea. Where this program and the Convention differ, the Convention is right and this is wrong.
2. **A lesson is not a licence.** Nothing here replaces a qualified course, a proper look-out, or the judgement Rule 2 requires of you.

3. **The drawings are schematic.** Real lights are dimmer, further apart in range than in angle, and often obscured by deck lighting, haze, shore lights and the motion of your own boat.
4. **Bearing drift is an observation, not a prediction.** It tells you what is happening now. It assumes both vessels hold course and speed, and the moment either alters, you start again.
5. **In restricted visibility you do not have a stand-on vessel.** Rule 19 is a different rule from Rule 15, and the lessons keep them apart.